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ORIGINAL ARTICLE

Cryptographic Data Security for Reliable Wireless Sensor Network

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Abstract Wireless Sensor Networks (WSNs) are widely utilized in a variety of applications. Attackers might interfere with equipment, and eavesdrop on conversations. Attackers can also modify the data which has been communicated or connect unauthorized devices to the network if such devices are deployed in the physical environment. To ensure WSN security, messages exchanged between communicating nodes must be encrypted, and for encryption and decryption, the network must store a key. Consequently, Key management is of significant importance in achieving security in WSN. Obtaining such a key arrangement in a resource-constrained environment is required for secure and energy-efficient data transmission in WSN. Current algorithms have certain security issues, such as vulnerability to plaintext attacks, brute force attacks, side-channel attacks, and computational complexity. Cryptography along with clustering has been found to be a practical and efficient technique to improve the secure and energy-efficient data transmission in WSNs. In this article we have used an asymmetric Elliptic Curve Cryptography (ECC) method for key generation whereas encryption and decryption of data, has been done by a hybrid of Advanced Encryption Standard (AES) and ECC cryptography A novel algorithm has been developed that incorporates both techniques (Advanced Encryption Standard (AES) and ECC cryptography) along with clustering through standard LEACH protocol for improving energy efficiency, data security, and network lifetime. The proposed algorithm has been able to overcome the issue of key exchange that plagues AES, simpler than ECC and more reliable than AES. More significantly, it is able to encounter a variety of security threats including side-channel attacks. The proposed algorithm is better and has been compared to current techniques with respect to time complexity, encryption time and decryption time to demonstrate its efficacy.

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1. Introduction

A wireless sensor network (WSN) can be used for analysis of temperature, humidity, vibrations, seismic activity and various other parameters. It consists of sensor nodes, spread over a topographical region. The energy required by the system to perform tasks is supplied by a power source. The problems with WSN's reliability are routing, versatility to deal with all types of data, power consumption and fault tolerance, quality of service, accessibility and protection. Our mission is to ensure safe and efficient data/information delivery to the desired destination in WSN with the best data aggregation using sufficient routing to achieve a high degree of security in WSN [1,2]. The four core security issues are data trustworthiness, data integrity, data authenticity and data availability. To provide security to the network's data communication channels, confidentiality and the authenticity of the data at each node are required [3]. WSNs are subject to attacks such as "Denial of Service (DoS)", "Sybil Assault", "Selective Forwarding", "Sink Hole attack", "Hello Flood assault", "Side-channel attack", "Brute Force attack", and "Node Capture attack" [4].

WSNs have limited resources for computation, power and memory. Clustering is one of the efficient approaches to provide energy efficiency in the WSN and with suitable cluster-head choice one can reduce the power consumption and network lifetime. Cluster formation in WSN is accomplished using a variety of clustering techniques. Effective cluster formation has the potential to significantly increase the lifetime of WSN.

LEACH (Low Energy Adaptive Clustering Hierarchy) has been proposed for routing in WSNs that forms cluster of nodes in a network. Cluster head (CH), which is a dedicated node with ample amount of energy in each cluster creates and manipulates the schedule of TDMA (Time division multiple access) and use CDMA (Code division multiple access) to transmit aggregated data from nodes to the BS where such data is required. Rests of the nodes are called as members of the clusters [5].

Although technology experts have explored a variety of methods for safeguarding data transfer, but data encryption is the most widely used and effective method. The data is "scrambled" by encryption so that it cannot be read by someone who is not intended to read it. Data encryption entails converting information into a form that only licensed individuals with a decryption key, will be able to read. Prior to encryption, data is referred to as plaintext, whereas after encryption, data is referred to as cypher text. Data encryption is done on purpose to protect confidential information while it is being stored or transferred from one system to another.

Advanced encryption algorithms have been proposed to start replacing or alter traditional encryption standards, and they play a major role in protecting data and data centers. Advanced encryption tools have features like data integrity, data authentication and successful data transmission that ensure secrecy. The provider of the data is recognized and verified through authentication; the integrity feature ensures that the information is not distorted, and the non-cancellation feature guarantees that the person who has sent the relevant data cannot halt the transmission. A few of the oldest data encryption involved changing letters in a sentence. The entire sentence became unreadable as a result, and it took a long time to figure out what the skewed characters meant. The decryption key, which allows it to read messages, is only available to the recipient party. People learned how to crack codes over time, so the encryption technique had to become more sophisticated to ensure that the message remained private.

Encryption methods include Asymmetric Encryption and Symmetric Encryption. One encrypts the data with a symmetric key, while the other encrypts the data with an asymmetric key. In symmetric-key cryptography, the data is encoded and decoded using the same key, whereas in asymmetric-key cryptography, the data is decoded and encoded using two distinct keys. Both mechanisms have their own advantages and disadvantages. The symmetric key technique has the advantage of being rapid and easy to use. A symmetric encryption method is low-cost, high-efficiency strategies for securing data and uses a single safe key and therefore it is less secure. Asymmetric algorithms, on the contrary, are sluggish and complicated, but they deliver a enhanced level of security. Two in-demand symmetric algorithms are Data Encryption Standard (DES) and Advanced Encryption Standard (AES) [6,7]. Asymmetric key cryptography, sometimes known as public-key cryptography, is a type of symmetric key encryption, which encrypts and decrypts messages using unique keys. RSA, Elliptic Curve Cryptography (ECC) based Elliptic Curve Digital Signature Algorithm (ECDSA) and the Elliptic Curve Digital Signature (ECDH) Algorithm is an example of asymmetric encryption methods. ECC is also used in Diffie Hellman [8–11]. As a result, combining symmetric and asymmetric encryption algorithms can be one of the best potential solutions for encryption. As a result, hybrid cryptography has been proposed in this work to ensure data security. Hybrid encryption is used in a variety of applications. Fig. 1 depicts the key exchange between the source and destination in symmetric and asymmetric encryption.

The proposed hybrid cryptography method has achieved the goal of ensuring data security and user authenticity along with energy efficiency through LEACH clustering. To secure data transmission in WSN, we have implemented a hybrid cryptographic schema based on ECC along with AES. And for ensuring energy efficiency and enhanced network lifetime, we have used LEACH clustering. The suggested method has a better speed of operation, level of security and a high level of energy efficiency.

This paper is organised as follows: Section 2 presents objective and novel contributions of proposed algorithm. Section 3 contains brief summary of relevant work for certain current procedures. Section 4 contains the description of proposed strategy and methodology along with detail of the proposed hybrid encryption algorithm. Sections 5 provide the outcome and discussion of the proposed algorithm for WSNs. Finally, Section 6 concludes this paper with a summary.

1.1. Objectives and contributions

Because of the WSN's limited resources and attacks, most researchers are forced to choose between energy efficient routing and data security while transmission. Despite several attempts to strike a trade-off between these two objectives, the dynamic nature of WSN and the restricted resources available to sensor nodes, makes the task of finding a safe

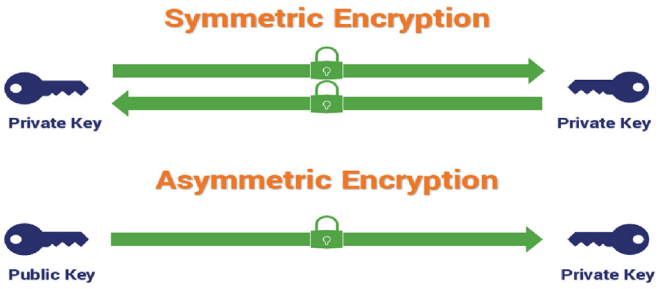


Fig. 1 Symmetric and asymmetric encryption.

and optimum network topology along with energy efficiency a difficult one. Many methods of trying to secure data transfer have been revealed by experts, but encryption process has been the most effective and simplest method that each and every user should be aware of and capable of using. Clustering on the other hand provides routing with more energy efficiency.

The purpose of this research is to create a hybrid cryptographic algorithm along with clustering. Objectives of the proposed and implemented work are as follows:

- Systematic review of security solutions for possible attacks in the wireless sensor network.
- Proposal of an algorithm for energy aware routing and secure transmission of using clustering protocol along with hybrid cryptography.

Weak security protocols can result in the loss of major, confidential and sensitive data in different wireless sensor network applications. Data loss that cannot be recovered may occur in a worst-case scenario due to security attack on information. The other consequences of weak security controls, such as high risk network and busy server congestion, loss of user confidence, and high cost of recovery. There are different problems and obstacles for WSN to develop a safe solution. We have reviewed basic terminology for sensor network security specifications, security risks, energy efficiency and security strategies based on cryptography and clustering. We have developed a security solution for the wireless sensor network. Random failure of nodes is also very likely in real-life implementation situations. Due to limitation of resources in the sensor nodes, conventional safety mechanisms with high overhead computing and connectivity are infeasible in WSNs. Therefore, the development and implementation of secure and energy efficient WSNs is an especially difficult task. We have developed a methodology for improving the energy efficiency and securing data in WSNs by utilizing the concept of hybrid cryptography and clustering based routing.

1.2. Literature review

Due to the limitless possible applications, wireless sensor networks are currently attracting considerable attention. Using various techniques to cope with the challenge of achieving reliability in wireless sensor networks, various protocols have been suggested. One of them is cryptography for data security. Security measures have been used to keep personal data secure while it is being stored or transferred through one system to the other. People have learned to crack commands over time, so encryption techniques must become even more intelligent to ensure that the message remains confidential. When the methodologies are used for transmissions, the information is first altered into an unreadable cypher text and sent in this format, after which the receiver decodes the cypher text back into its original format using a secret key or a password. If an intruder gains access to the file before it reaches the final computer, they will be unable to read it because it is encrypted.

The Triple Data Encryption Standard (Triple DES) algorithm employs three different block cypher algorithms to each data block. The size of the key is increased to give more security by improving the encryption capability. Each block is made up of 64 bits of data. Three keys are utilized in this encryption technique, each of which is 56 bits long [12]. The Blowfish Encryption Algorithm was once a serious contender for being the greatest encryption standard, but the current AES turned out to be the best one. This algorithm has a number of unique properties that sets it apart from other standards. This cryptographic protocol employs substitution-boxes, which are pre-computed and key-dependent Substitution-boxes (S-boxes). This means that, despite the presence of the S-box, the encrypted data is decrypted using the cypher key. The purpose of the S-box is to hide the key link between the encrypted text and the key. Also, the two-fish encryption standard is widely regarded as a safe option. Encryption protocols with keys of 128 bits or more are considered secure against attacks. The block size of two-fish is 128 bits. Due to its unique combination of design, robustness and speed, two-fish is considered the finest AES protocol. Today, the Advanced Encryption Technology (AES) is the most widely used symmetric encryption standard. A successor for DES was needed because of its tiny key size and limited processing capabilities, which led to the invention of AES. It has been proven to be more than six times quicker than Triple DES [13]. The ECC (Elliptic Curve Cryptography) method was implemented in 2004–05. The collection of points that satisfy the mathematical equation is known as an elliptic curve ($y^2 = x^3 + ax + b$) in the ECC encryption process. In ECC, a number indicating a curve point is multiplied by another number, yielding a new curve point. Determination of the new point on the curve is required in order to solve this puzzle. Even if you know the original position, it is very hard to find the new point using ECC's mathematics [14].

Authors proposed an algorithm using Diffie-Hellman's key exchange method for implementing the ECC cryptographic approach [15]. The sender encrypts the ECC and the unpredictable scheme with wide integers. Finally, from the accessible

encrypted large integers, the original results were obtained. Authors have shown that the proposed algorithm is secure and efficient as compared to all other current systems. A new lightweight encryption strategy has been proposed to improve network bandwidth and wireless sensor node computing sources [16]. To counter the threats, they used genetic operations (“mutation” and “crossover operations” and “XOR operations”) and a lightweight block cipher as a solution.

Secure Data Aggregation and Verification (SDAV) were proposed by the authors in [17]. The authors chose elliptic curve cryptography (ECC) over standard asymmetric methods because of its small key size and efficiency in terms of computation and bandwidth. Indeed, the ECC allows a single sensor to generate a signature while the base station handles the verification. In SDAV, the aggregator takes its members' encrypted data, decrypts it, averages it, and then provides the result to the members. When such disparity between the unit of measure and the mean received achieves a certain threshold, each participant creates a partial signature and sends it to the aggregator using the group key founded during the deployment phase. The Merkle hash tree has been used to ensure the authenticity of the measures if it is denied. The answer is the average aggregate function [18]. This method is inaccessible to eavesdropping. It is also resistant to Sybill attacks because each of the nodes has an identification number. This method, however, is vulnerable to attacks that deliver selected packets and replay attacks. Packets can be ignored by a compromise aggregator.

The authors of [19] presented a method for efficiently aggregating encrypted data. Data integrity and confidentiality are protected by the suggested approach. The fundamental concept is to utilise a simple modular addition instead of XOR operation). Passive attacks are not a problem with this method. Active-threat protection has not been explored in this approach. This system employs a probabilistic approach. The authors have offered strong hybrid cryptography and have demonstrated through outcomes that the proposed scheme provides greater security while requiring less encryption and decryption time. The authors used chaotic maps for data encryption and decryption after implementing the LEACH protocol for data routing. [20] Shows that chaotic maps can be broken using a variety of methods. A different study makes use of logistic and Kent chaotic maps [21]. The following are the primary drawbacks of the algorithms present in literature:

- In most of the proposed algorithms, there was only a finite amount of battery, and if even one node's battery drains, the entire WSN's data transmission was jeopardized.
- The decryption operation has been used in the algorithm which is not suitable for encryption of large data. If it is used to process huge amounts of data, the system becomes exceedingly sluggish. Even the integrity has been jeopardized [15].
- Authors in [22] have proposed secure data encryption, authentication as well as reliable end-to-end connectivity. Any event reporting could be disturbed, and communications could be read if an attacking player discovers a node's private key. Because of the inefficient processes used during the encryption step, the procedure is very vulnerable to attacks.
- According to previous research, the majority of the methodologies provide protection against multiple attacks. However, the majority of the proposed solutions do not combine routing and data security.

Elliptic Curve Cryptography arises as one of the most effective and convenient research methods for trying to secure WSN communication in the asymmetric cryptography category after investigating all of the existing options for safeguarding transmission of data in WSNs. One of the most often utilized techniques for protecting communication on nodes in WSNs is ECC. Due to the quickest methodology of multiplication in ECC, this cryptographic technology with its safe mechanism minimizes the computing cost. One of the most important features of ECC is its small key length, which reduces computational overhead. These are only a few of the important aspects that make ECC a better choice than RSA and DSA. ECC, on the other hand, provides similar levels of security as RSA but with significantly smaller operand sizes, such as 160–1204 bits versus 256–3072 bits and so on [23]. Due to its complete exponential duration, ECC delivers more strength per bit size than any other cryptosystem [24]. The fundamental signing and verification method of digital signatures utilizing an elliptic curve cryptographic technique is shown in Fig. 2 a, b. ECC uses shorter key lengths, which consumes less bandwidth and requires less memory to store keys. Furthermore, one of the benefits of ECC is that it allows users to pick the curve on a regular basis. This essential aspect of ECC aids in improving the network's security level for data transfer. ECC results in safe and efficient network. Based on analysis of the previous research undertaken, the key sizes of three algorithms RSA, DSA and ECC that have displayed the same level of security has been shown in Table 1.

AES is single-key symmetric key encryption and decryption techniques. Three block cyphers are included in AES: “AES-128”, “AES-192” and “AES-256”. AES is a cryptographic algorithm that is used to encrypt sensitive data in software and hardware all around the world. AES was developed to replace Data Encryption Standard (DES), which was vulnerable to brute-force attacks. It provides sufficient security against a range of attacks.

The AES design is shown in Fig. 3. Encryption and decryption of a block of messages, AES-128 employs a 128-bit key length, while AES-192 and AES-256 use 192-bit and 256-bit keys, respectively. Using 128, 192, or 256-bit cryptographic keys, each cypher encrypts and decrypts data in 128-bit blocks. There are 10 rounds for 128-bit keys, 12 rounds for 192-bit keys, and 14 rounds for 256-bit keys. Substitution, transposition, and mixing are just a few of the many processing steps that turn plaintext into cypher text in a round.

Security analysts claim AES is resistant to brute-force attacks, which require testing of all key combinations possibilities until the correct one is found. Nevertheless, the key lengths used for encryption must be adequate to prevent modern machines from cracking it, even if proven technology has improved due to Moore's law..

In several ways, AES is preferable. AES is a high speed encryption algorithm that has been used in both software and hardware implementations. It is based on the substitution-permutation network architecture concept. The finite element model is used for the majority of AES estima-

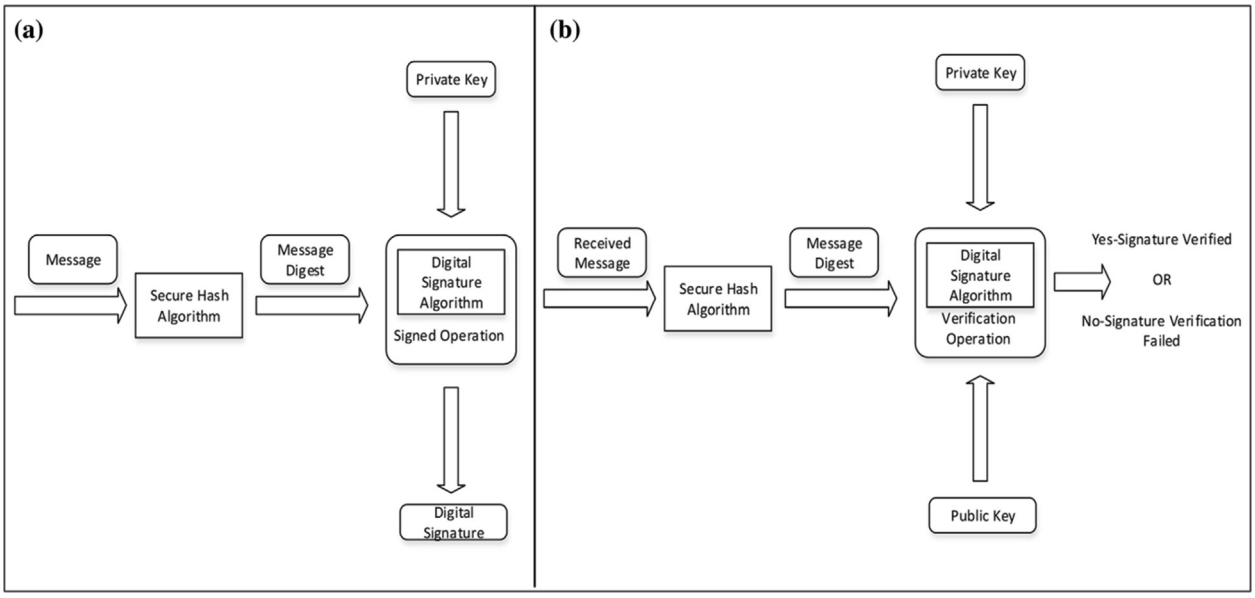


Fig. 2 (a) Elliptic curve digital signature algorithm signing mechanism. (b) Elliptic curve digital signature algorithm verification mechanism.

Table 1 Comparison between RSA, DSA and ECC.

Bits	ECC	DSA	RSA
80	160–223	1024	1024
112	224–225	2048	2048
128	256–383	3072	3072
192	384–511	7680	7680
256	512 +	15,360	15,360

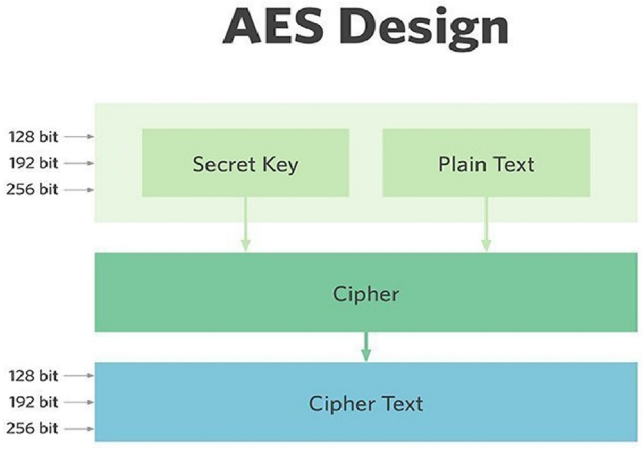


Fig. 3 Relationships between the Secret Key, Plaintext, Cipher, and Cipher Text are depicted in visual diagrams.

tions. The number of conversion rounds necessary to turn the input, known as plaintext, into the final output, known as cypher text, is determined by the key size used for an AES cypher. AES encryption and decryption software is widely available. It's used when data is so sensitive that only authorized individuals should have access to it. The AES name, which is the world's most widely acknowledged encryption standard, continues cropping up on all computer screens when it comes to cyber security. The AES standard is made up of three block cyphers, each of which utilizes cryptographic keys to encrypt and decode data in a 128-bit block. The key sizes are deemed suitable for securing classified data to a reasonable level of secrecy [26].

In [25], the authors used ECC for encrypting the plain text applied Dual RSA on the plain text's hash value. It was about a four times as quick as conventional RSA and used less amount of memory. The process of encrypting the key was time consuming. The plain text was encrypted using AES and ECC in order, followed by MD5 hash value computation and plain text hash calculation. Because of the sequential method, this technique requires less memory but takes longer to execute [27]. The DES method was utilized for data transfer in the next algorithm, and the RSA algorithm was employed to encrypt the DES key. It is more secure and has a better efficiency [28]. Authors in [29] utilized symmetric cypher (AES-Rijndael) for encryption and public key cryptography (RSA) for authentication using the hash function SHA-512 for more secure transmission.

With asymmetric cryptography, performance will suffer owing to limited memory, energy, and computing resources after conducting a literature analysis of various research publications. Whereas with symmetric cryptography, data transit is less safe since if the algorithm's key is disclosed, the entire system's security is jeopardized. For example, if the algorithm's key is compromised, the entire system's security is jeopardised. As a result, we should develop an algorithm that is both secure and simple enough to provide appropriate data protection. Proposed algorithm should ensure data integrity and confidentiality. As a result, we created an algorithm that is provides data security while maintaining energy efficiency via clustering-based routing. We used the ECC method for

key generation, and AES hybrid ECC and LEACH-based clustering for data encryption and decryption.

The methods involved in developing the overall reliability strategy has been explored and the best possible combination of the available security options has been discovered in WSNs. Wireless sensor network protection and communication are an integral property of wireless sensor networks (WSNs). Security tests the efficiency of a WSN's monitoring, while connectivity allows the sensor to supply the sensed data to the sink.

1.3. Research design & methodology

Symmetric with asymmetric encryption, symmetric with another symmetric encryption, and asymmetric with another asymmetric encryption are all examples of hybrid encryption. Hybrid encryption combines the best of both approaches to create a synergy that allows for the development of secure encryption systems. The symmetric encryption approach is ideal for encrypting huge amount of data quickly. Nonetheless, it lacks identification and verification, which is essential for security. Asymmetric encryption with a public/private key combination, on the other hand, ensures that the data is only accessible by the intended recipient. When used for large group of data, however, this verification slows down the encryption process. In many applications, it was necessary to encrypt data at a rapid rate which can be done by symmetric encryption while simultaneously requiring identity verification to guarantee that users were conversing with the right organization which can be achieved by asymmetric encryption.

For efficient network management, the proposed scheme employs a clustering approach. Furthermore, the Cluster Head (CH) is in charge of major network decisions. The proposed method is advantageous and highly secure for the critical WSN environment. Furthermore, in the proposed scheme, we built a cluster of sensor nodes. The LEACH protocol is then used to dynamically select a CH for each cluster.

Block diagram present in Fig. 4 shows the research design and methodology for hybrid cryptography and clustering based secure and energy efficient data transmission. After conducting literature survey we found that hybrid cryptography can be used as the suitable approach to prevent the network from various security attacks (Man in the middle, Sybil, black hole, side channel and so on).

We have attempted to overcome the constraints found in earlier work in the area of secure data communication in WSNs by various authors. With the goal of ensuring high security, we have proposed an algorithm that secures WSN.

*1.4. Proposed **algorithm***

In the proposed algorithm, for routing, LEACH protocol has been used and data has been encrypted using a hybrid method using ECC as well as AES for secure and efficient encryption. Decryption takes place using ECC. The proposed algorithm has two main steps which have been explained in details in this section.

The nodes in LEACH form local clusters with each cluster having cluster's head to gather information from all cluster members to perform processing steps on the data, and send it to the remote base station. Therefore, a cluster-head node consumes far more power than a non-cluster-head node. If the cluster-head sensor nodes were fixed throughout the system lifespan, similar to static clustering algorithm, then they would quickly exhaust their limited energy. It becomes inoperable when the cluster-energy head's energy runs out. As a result, when a cluster-head node dies (for example, by depleting all of its battery energy), all of the cluster's nodes lose communication capability. To overcome this, in LEACH there is a rotation of the cluster-head position among the cluster members in order to avoid load on any single sensor in the network. The cluster-head node can build a TDMA schedule that specifies when each node must transmit data because it is aware of all cluster members. All internal modules are disabled, allowing the nodes to sleep for as long as they can. Intra-cluster collisions are also avoided by using a TDMA data transfer plan.

The proposed algorithm's operation is separated into cycles. The rounds starts in a phase to arrange clusters as shown in Figs. 5, 6, 7, and 8, and several blocks of data are transmitted while maintaining safety through hybrid cryptography from the nodes to the cluster-head and on to the base station in a steady-state phase. The time-synchronization of all nodes is necessary to begin the setup phase at the same time. The steady-state phase is time taking than the set-up phase in order to reduce set-up overhead. The proposed algorithm's steps are as follows:

STEP 1: Self-Configuring Cluster Formation

LEACH employs a distributed algorithm to build clusters in which nodes have no centralised control and they decide in an autonomous way to eliminate communication with the remotely located base station and form distributed clusters without knowing where any of the network's nodes are located. Furthermore, no global communication is necessary for cluster creation. A good cluster formation exclusively through local decisions made by each node independently can be done.

Set-up Phase

In this phase after election as a cluster-heads, these cluster head nodes send the notification to all other nodes in the network by broadcasting an advertisement message (ADV) by utilising the concept of non-persistent carrier-sense multiple access (CSMA) MAC protocol, that they have been elected as cluster-heads for the current round. This ADV has the ID of the nodes to be identified by a header as an announcement message. This ADV should reach all the nodes of the cluster to avoid hidden terminal problem and it virtually eliminates collisions while using CSMA.

Some nodes on the network's edge may not receive any announcements because of low energy required by the ADV to reach, and could made them unable to participate in current round of the algorithm. The ADV message has been kept small in size in order to not utilize much power to send them and the transmit power has been kept high enough so that the ADV message can reach efficiently by all nodes in the network. The strength of the received signal reached to non-cluster-head nodes by cluster-head via ADV message helps them decide that they belong to which cluster head by utilizing least amount of communication energy.

The channel has been assumed as symmetric for propagation of signal and the cluster-head ADV message with the highest signal strength is the cluster-head for which the highest amount of transmitted energy is required for communication. In case of interference in the channel for communication between two physically close nodes, the communication with

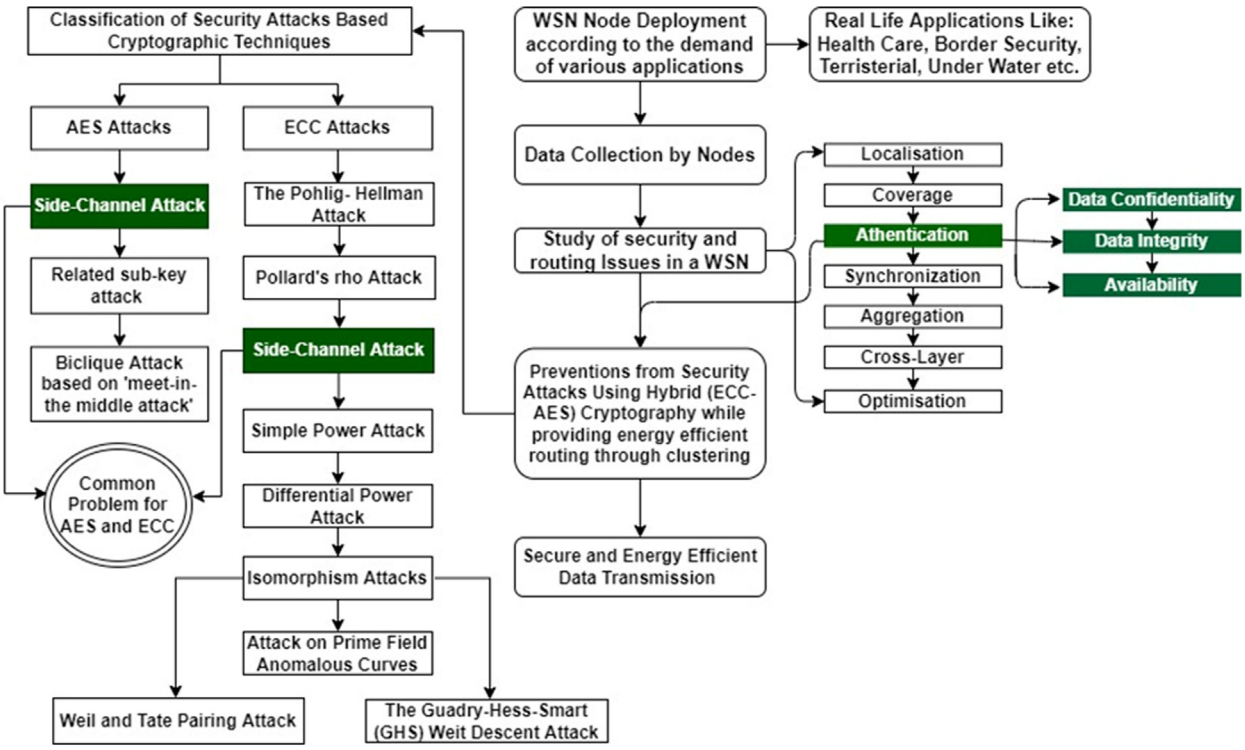


Fig. 4 Block Diagram of Research Design and Methodology.

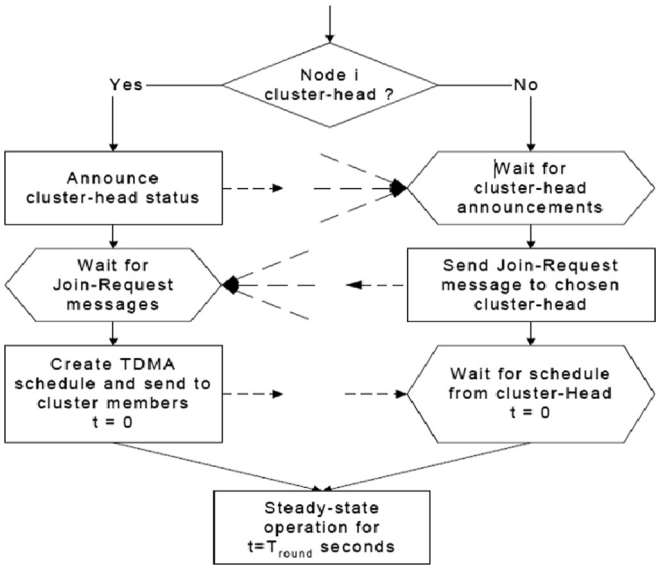


Fig. 5 Flow-graph of the distributed cluster formation algorithm for LEACH.

another cluster-head located farther away becomes easier, and the sensor would choose the cluster-head that is spatially farther away but "closer" in a communication sense. In the event of a tie, a cluster-head is chosen at random.

After the decision of cluster formation along with their cluster heads, the cluster members send a join-request message (Join-REQ) which follows a CSMA MAC protocol to the respective cluster head that they will be a member of the that cluster and the message has been kept small containing only the node's ID, cluster's ID, head's ID, and a header. Now the cluster members have idea of how much energy is required to send Join-REQ message required towards the cluster-head based on the received power of the advertisement message and can adjusts its transmit power accordingly.

To avoid collision between messages in LEACH and to allow the radio components of each non-cluster-head node

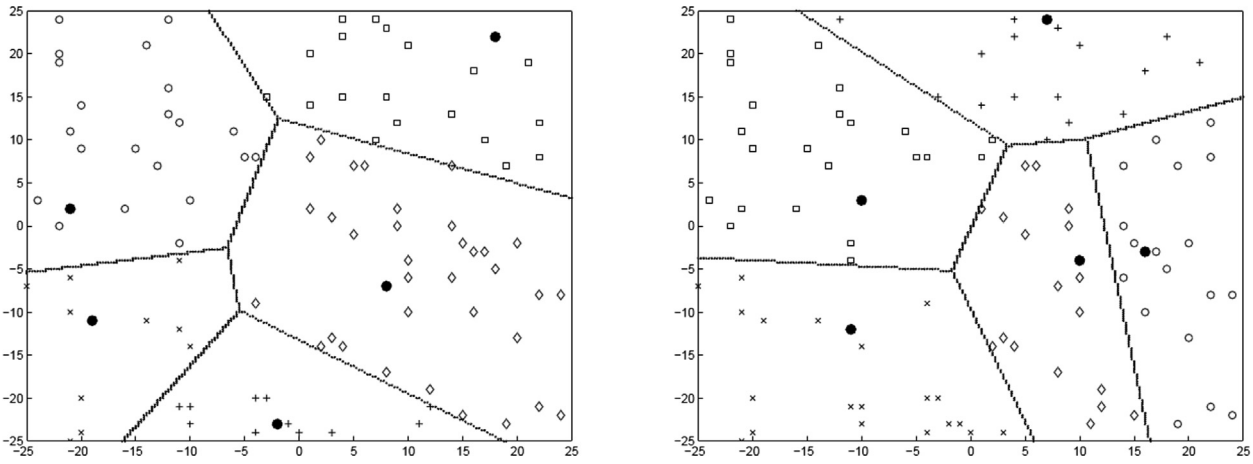


Fig. 6 In two different rounds of LEACH, dynamic cluster formation was seen. The cluster-head nodes are identified with and all nodes designated with a given symbol belong to the same cluster •.

to be turned off at all times except when transmitting, cluster heads follows a TDMA schedule, which reduces the energy dissipated by the individual sensor nodes. After making cluster members aware of the TDMA schedule, the set-up phase completes and the steady-state operation can begin. The flowgraph of our distributed cluster generation algorithm is shown in Figs. 5 and 6.

Step 2: Hybrid Cryptography Integrated Steady-state Phase

In this step, first the nodes will send data to the cluster-head only once per frame during their assigned transmission slot because each slot in which a node transmits data is constant and the time required for a frame of data transfer is proportional to the number of nodes in the cluster. The distributed algorithm which determines the cluster-head nodes only give us the expected number of clusters per round (k), but it does not guarantee that k clusters are present at each round. And the set-up phase too does not guarantee that nodes are distributed evenly among cluster-head nodes. So as a conclusion, the number of nodes per cluster in LEACH is highly variable, and the amount of data each node can send to the cluster-head varies with the number of nodes in the cluster. To receive all data from the cluster's nodes, the cluster-head must keep its receiver turned on.

Figs. 8 and 9 show how the proposed hybrid cryptography was applied to that data to ensure data security. The steps taken by the process depicted in Fig. 8 are as follows:

- Cluster head (CH) generates the Key by ECC asymmetric encryption (Encrypt_ECC (msg, pub_{key})) for the message recipient.
- To return the message (msg) to its original form, the generated key is utilized to encrypt and decrypt it using a hybrid encryption approach combining asymmetric ECC and symmetric AES.
- CH then encrypts the received data from CH members along with its calculated hash function using 256-bit AES secret key (pub_{key}).
- Base station (BS) then decrypts the received data and recovers them by symmetric decryption method with the help of private key ($priv_{key}$) so that the shared key for the whole network can be communicated securely.

2. Encryption process

The plaintext is divided into n blocks $Binary_i$ during encryption. There are 256 bits in each block. Then it's split into two sections: Section 1- msg_i (0: num/2–1) blocks and section 2- msg_i (num/2: num-1).

As the ECC algorithm is the most secure public key algorithm, it is utilised to protect secret keys. ECC can solve fully exponentially problems rather than sub-exponentially for other public key systems, ECC requires a smaller key size than other methods, which means it requires less amount of memory [1]. This algorithm permits communication nodes to accommodate a greater number of requests with less number of packets being dropped. Because ECC costs more power than symmetric algorithms, AES minimizes power consumption while improving system performance [2]. We can save power and obtain speeds of up to 25 % for encryption and nearly 20 % for decryption when we use AES with ECC [3].

The key key_i, which is the 256-bit secret key of the AES encryption technique, is used to encrypt msg_i using AES. $E_{encrypts}$_ECC key_i, resulting in Key_j with length L.

$$msg_i = \sum_{i=0}^{i=\frac{num}{2}-1} (Binary_i)\,\mathrm{num}/2 \leq \mathrm{i} \leq \mathrm{n} - 1$$

$$Key_i = Encrypt_{ECC}\left(pub_{Key_i}, Key_{i-1}\right) 0 \leq \mathrm{j} \leq \mathrm{L} - 1$$

Where $E_{encrypt}$_ECC is Elliptic Curve encryption function. It encrypts the data with the trust centre public key (pub_{key}), which is then used to authenticate the key.

$$CT_i = E_{AES}\,(Key_j, Binary_i)$$

$$msg_i = \sum_{i=0}^{i=num-1} (Binary_i)\mathrm{num}/2 \leq \mathrm{i} \leq \mathrm{n} - 1$$

P_{num} and q_{num} are the two randomly selected prime number in this algorithm. Then $\times = P_{num}{*}q_{num}$, $Ø(x) = (P_{num}\text{-}1) * (q_{num}\text{-}1)$. A number relatively prime to Ø is chosen; d. Then, e is calculated such that $e \times d = 1 \bmod \varphi(x)$, and $Public_{key}$ (e, x) is used for encryption.

$$S_i = (Binary_i^e) mod x$$

ASCII for $Binary_i$ is get and converted to binary

$$F_i = \mathrm{ASCII}\, Binary_i$$

where F_i is a function used to convert message block to ASCII. S_i is a ciphered text using SHA-256.

$$CT_i = (S_i \oplus \boldsymbol{F_i})$$

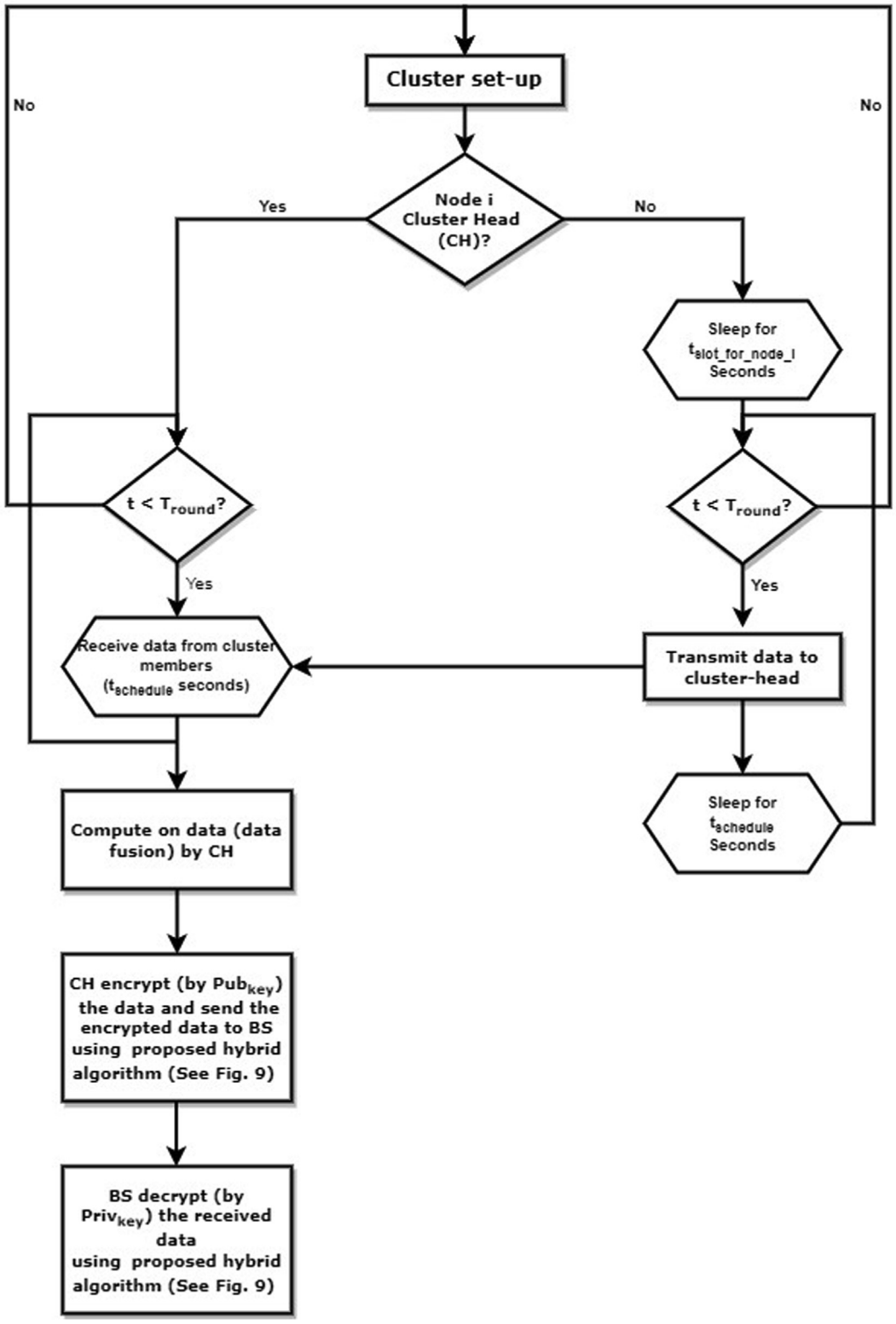


Fig. 7 Flow-graph for hybrid cryptography integrated steady state phase in LEACH.

SHA-256 is applied to the cipher texts ct_i and CT_i.

$d_i = \text{SHA256}(ct_i)$

$D_i = \text{SHA256}(CT_i)$

The two n/2 blocks are combined at the end of the encryption process to form cypher text of num blocks, which is then sent to the sink node. The respective hash values (di and Di)

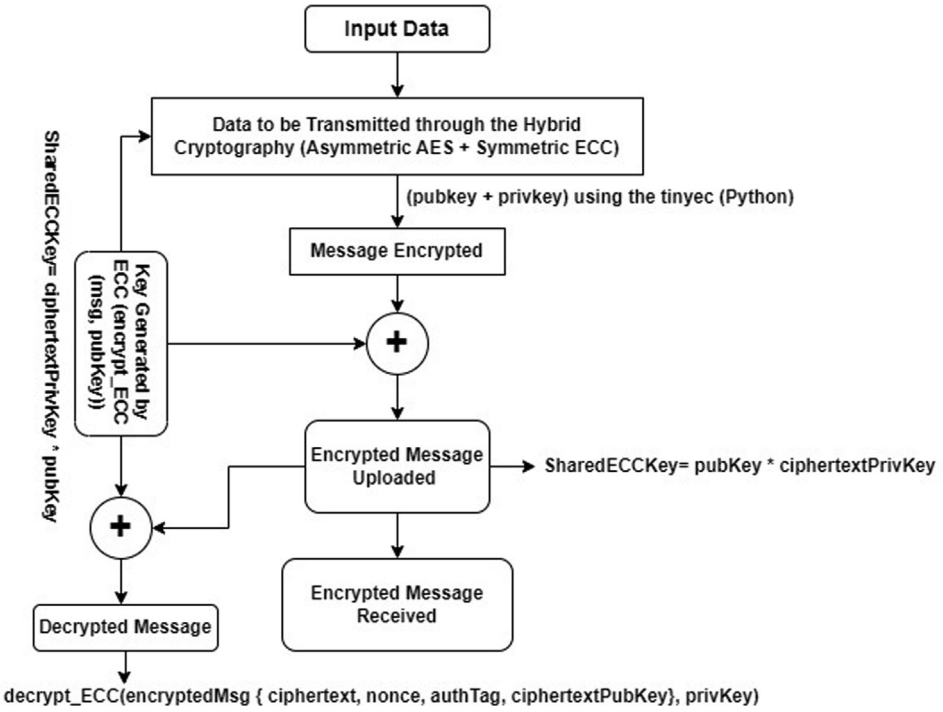


Fig. 8 Block Diagram of Hybrid ECC + AES.

with a size of 256 bits are concatenated and sent to the sink node simultaneously.

2.1. Decryption *process*

$ct_i = \sum_{i=0}^{i=\frac{num}{2}-1}(Binary_i) 0 \leq i \leq num/2 - 1$

$Key_i = Decrypt_{ECC}\left(pub_{Key_i}, Key_{i-1}\right) 0 \leq j \leq num/L - 1,$

$0 \leq j \leq L - 1$

ECC decrypts the AES key_j key, which has a length of L bits, to yield ki, which is utilised to decrypt the encryption.

$msg_i = D_{decrypt}AES\ (key_j, ct_i)$

$CT_i = \sum_{i=num/2}^{i=num-1}(Binary_i) num/2 \leq i \leq num - 1$

For decryption, private key is used. Following parameters are computed for the decryption process.

$d_{pnum} = d \bmod (p_{num} - 1)$, $d_{qnum} = d \bmod (q_{num} - 1)$, $S_{pnumi} = Si\ d_{pnum} \bmod p_{num}$, $S_{qnumi} = Si\ d_{qnum} \bmod q_{num}$,

$shared_{ECC} = (priv_{key} * CT_i)$

$PT_i = D_{decrypt}_AES_GCM\ (key_j, ct_i)$

The two num/2 blocks are combined at the end of the decryption procedure to create plain text (PT_i) for one block.

To generate an ECC asymmetric encryption key pair for the message recipient, the proposed technique was implemented using the tinyec library of the python framework known as pycharm. These keys will be used to encrypt and decrypt the message (msg) to return it to its original form using a hybrid encryption strategy of asymmetric ECC and symmetric AES. Following this, the cluster-head encrypts the message (msg) with the public key (pubKey), yielding the concatenated output cypher text (ciphertext), random AES initialization vector (nonce), authentication tag (authTag), and public key (ciphertextPubKey). The ciphertext, nonce, and authTag are obtained using symmetric AES-GCM encryption.

After generating an ephemeral ECC key-pair for the ciphertext, the encrypted message with public key (ECC (msg, pubKey)) function calculates the symmetric encryption shared key (sharedECCKey = ciphertextPrivKey * pubKey) of ECC. Because this is an ECC point, hashing the x and y coordinates of the point converts it to a 256-bit AES secret key (integer).

Cipher text is converted into a public key (ciphertextPubKey), which is a randomly generated ephemeral public key. This key will then be included in the encrypted message and will be used to decrypt the AES symmetric key. To encrypt and decrypt the encrypted message, data can be generated during encryption using a public key (ciphertext, nonce, authTag, ciphertextPubKey) and during decryption using a private key (privateKey). As a result, the plaintext message will be decrypted.

Finally, the AES-256-GCM cypher (from pycryptodome) encrypts the message and the concatenated output of ciphertext, nonce, and authTag using the 256-bit shared secret key (secretKey). The decrypted ECC key (encryptedMsg ciphertext, nonce, authTag, ciphertextPubKey, privKey) and its function (sharedECCKey = privKey * ciphertextPubKey) are used to calculate the shared ECC key for symmetric encryption. Because it is an ECC point, hashing the × and y coordinates first converts it to a 256-bit AES secret key. The shared secret key has a length of 256 bits. Using the secret key, the AES-256-GCM cypher is then used to decrypt the concatenation of ciphertext, nonce, and authTag (secretKey). In the case of an incorrect decryption key or an unmatched authentication tag, the output is the original plaintext message or an exception (authTag).

To encrypt and decode data, we used the ECC hybrid AES technique. They are among the most effective asymmetric and

Hybrid (ECC and AES) algorithm

1. *Import → registry from tinyec*
2. *Import → AES from Crypto.Cipher*
3. *Import → hashlib, secrets, binascii*
4. *Def → E_{ncrypt}_AES_GCM (msg, $secret_{Key}$)*
   aesCipher = AES.new ($secret_{Key}$, AES.MODE_GCM)
   Ciphertext, $auth_{Tag}$ = aesCipher. E_{ncrypt} _and_digest (msg)
   Return → (ciphertext, aesCipher, nonce, $auth_{Tag}$, $secret_{Key}$)
5. *Def → D_{ecrypt}_AES_GCM (ciphertext, nonce, authTag, $secret_{Key}$)*
   aesCipher = AES.new ($secret_{Key}$, AES.MODE_GCM, nonce)
   plaintext = aesCipher.decrypt_and_verify (ciphertext, $auth_{Tag}$)
   Return → plaintext
6. *Def → ecc_point_to_256_bit_key (point)*
   SHA = hashlib.SHA256 (int.to_bytes (point.x, 32, 'big'))
   SHA.update (int.to_bytes (point.x, 32, 'big'))
   Return → SHA.digest ()
7. *Curve = registry.get_curve*
8. *Def → E_{ncrypt}_ECC (msg, pub_{key})*
   $ciphertext_{PubKey}$ = secrets.randbelow (curve.field.n)
   *$shared_{ECCKey}$ = $ciphertext_{PubKey}$ * pub_{key}*
   $secret_{Key}$ = ecc_point_to_256_bit_key ($shared_{ECCKey}$)
   ciphertext, nonce, $auth_{Tag}$ = encrypt_AES_GCM (msg, $shared_{ECCKey}$)
   *$ciphertext_{PubKey}$ = $ciphertext_{PubKey}$*curve.g*
   Return → (ciphertext, nonce, $auth_{Tag}$, $ciphertext_{PubKey}$)
9. *Def → D_{ecrypt} _ECC (encryptedMsg, $priv_{Key}$)*
   (ciphertext, nonce, $auth_{Tag}$, $ciphertext_{PubKey}$) = encryptedMsg
   *$shared_{ECCKey}$ = $priv_{Key}$ * $ciphertext_{PubKey}$*
   $secret_{Key}$ = ecc_point_to_256_bit_key ($shared_{ECCKey}$)
   plaintext = D_{ecrypt} _AES_GCM (ciphertext, nonce, $auth_{Tag}$, $secret_{Key}$)
   Return → plaintext
10. *msg = b'Text to be encrypted by ECC public key and '\b'decrypted by its corresponding ECC private key'*
11. *print → original msg, msg*
12. *$priv_{Key}$ → secrets.randbelow (curve.field.n)*
13. *pub_{key} → $priv_{Key}$ * curve.g*
14. *encrptedMsg = E_{ncrypt}_ECC (msg, pub_{key})*
15. *encryptedMsgObj = {*
    'ciphertext': binascii.hexlify(encryptedMsg[0]),
    'nonce': binascii.hexlify(encryptedMsg[1]),
    'authTag': binascii.hexlify(encryptedMsg[2]),
    'ciphertextPubKey': hex(encryptedMsg[3].x) + hex(encryptedMsg[3].y % 2)[2:]
    }
16. *print → encryptedMsg, encryptedMsgObj*
17. *decryptedMsg = D_{ecrypt}_ECC (encryptedMsg, $priv_{Key}$)*

Fig. 9 Proposed Hybrid Algorithm.

symmetric key algorithms available in the literature. The time complexity and spatial complexity of AES are of the order of $O(1)$. Because ECC is a block cypher, its complexity is constant because it operates on a fixed length data, i.e. 128 bits, for a fixed number of steps. We used ECC for key generation and sharing (elliptic curve cryptography). The time complexity of ECC key creation is also constant, as the public and private keys of the sender and receiver are generated in the same amount of time complexity i.e. $O(1)$. ECC has a far lower space complexity than the second most used asymmetric key technique, RSA. As a result, a hybrid encryption method for secure data communication in WSN has been developed.

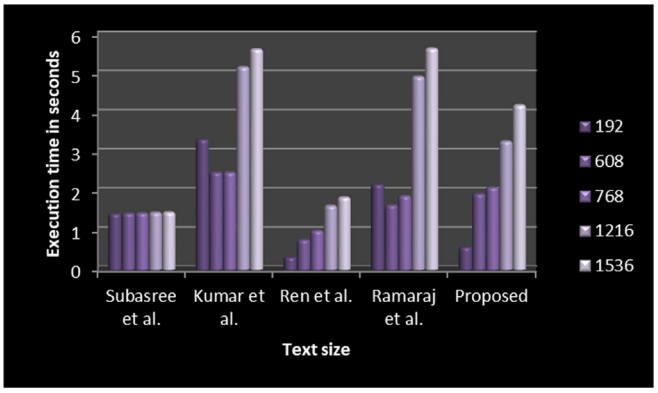


Fig. 10 Execution Time in ECC-AES along with other existing algorithms.

2.2. Results and discussions of AES and ECC-AES: a critical analysis

Algorithms [28,29,30,31] and the proposed method have been compared on text sizes of 192, 608, 768, 1216, and 1536 bytes. These algorithms have been compared for each text size so that the fastest algorithm can be determined. The entire execution time of the above-mentioned algorithms is displayed in Fig. 10, and it can be seen that [28,29,30,31] algorithms are slower than the proposed one in terms of execution time.

Another test has been performed to compare these methods. To begin with, text data was collected and a key of size 128 was specified. On this text data, the AES technique was used to encrypt it. The length of time it took to complete the encryption procedure was recorded. After encryption, the original text data was converted to cipher text data. The encrypted (cipher text) data was then decrypted to reveal the original text contents. This decoding process took even more amount of time.

Similar tests were carried out using a model called Hybrid ECC-AES Encryption and Decryption, which used a combination of the AES and ECC algorithms. The time taken for encryption and decryption operations was recorded using the same text material. Figs. 11 and 12 shows graphs representing the decryption time and encryption time taken by the existing and the proposed hybrid algorithm respectively. ECC-AES takes a comparable amount of time. ECC-AES takes a comparable amount of time here as well. The key sizes mentioned earlier were used to evaluate existing and the proposed Hybrid ECC-AES models for text data. The ECC-AES encryption time was recorded as 0.971 milli seconds using the 192-byte key. The AES encryption time for the 128-bit key was 3.74 s, whereas for the ECC-AES encryption time was 2.45 s. The AES encryption time for the 192-bit key was 3.67 s, while the ECC-AES encryption time was 2.47 s. AES encryption time was recorded as 3.75 while ECC-AES encryption time was recorded as 2.5 for the 256-bit key. For ECC-AES, decryption time was reported as 0.5739 milli seconds for utilizing the 192-byte key. The AES algorithm does not operate with the 64-bit key. AES decryption time was recorded as 2.82 for the 128-bit key. The ECC-AES decryption time was found to be 1.66. When it comes to the 192-bit key, the ECC-AES decryption time was 2.83 and the AES time was 1.69. All the time values are in milli seconds. The AES time decryption for the 256-bit key was 2.85 was recorded, while ECC-AES decryption time was 1.71.

The hybrid ECC-AES model was found to take less amount of time than the AES model and other existing models. The results clearly demonstrated that the hybrid ECC-AES approach takes less time to encrypt and decrypt data than the traditional AES approach. From the overall evaluation of selected cryptographic algorithms, it has been observed that the proposed hybrid (AES-ECC) algorithm is more reliable than the others in terms of security and energy efficiency. Proposed hybrid technique provides a faster as well as secure procedure of both encrypting and decrypting data as compared to

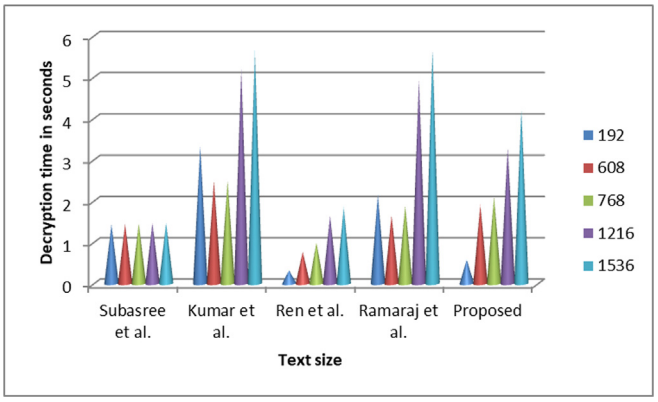


Fig. 11 Decryption time Of ECC-AES along with other Existing Algorithms.

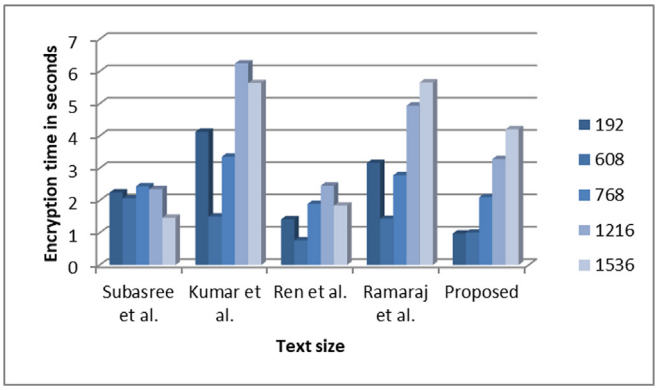


Fig. 12 Encryption time of ECC-AES Along with other Existing Algorithms.

the other models and jointly secure the data from various attack including side channel attack, brute force attack and various other attacks [32].

3. Conclusion

In the research community the area of wireless and mobile computing, during the past decade, wireless sensor networks (WSNs) have gained a lot of interest. There are innumerable growing applications of WSN, ranging from home and office indoor deployment situations to outdoor deployment in the adversary's territory in a military battlefield. These networks, however, are vulnerable to various security threats, because of their deployment in remote areas. This may adversely affect their performance. In this article we have investigated WSN security issues and security risks along with LEACH clustering for energy efficient routing. In addition, we have recognized a series of effective countermeasures that can be used to enforce their proposed guidelines. In this respect, the use of technology called cryptography to solve such security problems has been demonstrated and their shortcomings have been reduced for making it more usable for WSNs. By understanding the effect of data security attacks in WSNs, we have implemented algorithms to decrease the challenges of WSN security using hybrid cryptography.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

[1] K.N. Qureshi, A.H. Abdullah, Adaptation of wireless sensor network in industries and their architecture, standards and applications, World Appl. Sci. J. 30 (10) (2014) 1218–1223.

[2] V. Kumar, A. Jain, P. Barwal, Wireless sensor networks: security issues, challenges and solutions, Int. J. Inf. Comput. Technol 4 (8) (2014) 859–868.

[3] D. Kumar, S. Chand, B. Kumar, Cryptanalysis and improvement of an authentication protocol for wireless sensor networks applications like safety monitoring in coal mines, J. Ambient Intell. Humaniz. Comput. 10 (2) (2019) 641–660.

[4] P.B. Hari, S.N. Singh, Security issues in wireless sensor networks: current research and challenges, in: Paper presented at the 2016 international conference on advances in computing, communication, & automation (ICACCA) (Spring), 2016.

[5] W. Heinzelman,"Application Specific Protocol Architecture for Wireless Sensor Networks", B.S., Cornell University 1995, M.S. Massachusetts Institute of Technology (1997) at MASSACHUSETTS INSTITUTE OF TECHNOLOGY (MIT) June 2000©.

[6] G. Singh, Supriya,, A study of encryption algorithms (RSA, DES, 3DES and AES) for information security, Int. J. Comput. Appl. 67 (19) (2013) 33–38.

[7] W. Burr, Selecting the advanced encryption standard, IEEE Secur. Priv. 1 (2) (2003) 43–52.

[8] M. Frunza, Gh. Asachi, Improved RSA encryption algorithm for increased security of wireless networks, in: ISSCS International Symposium,vol. 2, 2007.

[9] R. Kodali, N. Sarma, Energy efficient ECC encryption using ECDH, Lecture Notes in Electrical Engineering, vol. 248, Springer, 2013, pp. 471–478.

[10] D. Johnson, A. Menezes, S. Vanstone, The elliptic curve digital signature algorithm (ECDSA), Int. J. Inf. Secur. 1 (1) (2001) 36–63.

[11] M. Balitanas, WiFi protected access-pre-shared key hybrid algorithm. Int. J. Adv. Sci. Technol., 2009, 12.

[12] S. Karthik, A. Muruganandam, Data encryption and decryption by using triple DES and performance analysis of crypto system, Int. J. Scient. Eng. Res. (IJSER), ISSN (Online) 2 (11) (2014) 2347–3878.

[13] J.D. Gaur, A. Kumar Singh, N.P. Singh, G.V. Rajan, "Comparative study on different encryption and decryption algorithm, in: 2021 International Conference on Advance Computing and Innovative Technologies in Engineering (ICACITE), 2021, pp. 903-908, doi: 10.1109/ICACITE51222.2021.9404734.

[14] S.K. Verma, D.B. Dr, Ojha, A discussion on elliptic curve cryptography and its applications, JCSI Int. J. Comput. Sci. Issues Vol. 9, Issue 1, No 1 (2012), ISSN (Online): 1694–0814.

[15] X. Zhang, X. Wang, Digital image encryption algorithm based on elliptic curve public cryptosystem, IEEE Access 6 (2018) 70025–70034.

[16] A. Dutta, K. Naveen Kumar, N. Sai, R.R. Chintala, An efficient light weight cryptography algorithm scheme for WSN devices using chaotic map and GE, Int. J. Pure Appl. Math. 118 (20) (2018) 861–875.
[17] S. Ozdemir, H. Ichakawa et al. (Eds.), Secure and reliable data aggregation for wireless sensor networks, LNCS 4836, pp. 102–109 (2009).
[18] Z. Dawahdeh, N.S. Yaakob, A new modification for menezes-vanstone elliptic curve cryptosystem, J. Theor. Appl. Inf. Technol. 85 (2016) 290–297.
[19] elena Misic, Vojislav Misic,, Wireless personal area networks performance interconnections and security with, John Wiley & Sons Ltd, 2008, IEEE 802.15.4.,.
[20] M.P. Nidarsh, M.G.P. Devi, Chaos based secured communication in energy efficient wireless sensor networks, Chaos 5 (6) (2018) 742–747.
[21] M.I. Sobhy, A.-E. Shehata, Methods of attacking chaotic encryption and countermeasures, in: 2001 IEEE international conference on acoustics, speech, and signal processing, Salt Lake City, UT, 7–11 May 2001, pp.1001–1004. New York: IEEE.
[22] M. Panda, Security in wireless sensor network using cryptography techniques, Am. J. Eng. Res. (AJER) 3 (2014) 50–56.
[23] M. Rajput, U. Ghawte, Security challenges in wireless sensor networks, Int. J. Comput. Appl. 168 (2017) 24–29.
[24] H. Yetgin, K.T.K. Cheung, M. El-Hajjar, L.H. Hanzo, A survey of network lifetime maximization techniques in wireless sensor networks, IEEE Commun. Surv. Tutor. 19 (2) (2017) 828–854.
[25] M. Pitchaiah, Philemon Daniel, Praveen, Implementation of advanced encryption standard algorithm, Int. J. Scient. Eng. Res.Volume 3, Issue 3, March -2012, ISSN 2229-5518
[26] A. Gyrard, M. Serrano, Connected Smart Cities: Interoperability with SEG 3.0 for the Internet of Things, in: 2016 30th International Conference on Advanced Information Networking and Applications Workshops (WAINA), 2016, pp. 796-802, doi: 10.1109/WAINA.2016.151.
[27] S. De, P. Barnaghi, M. Bauer, S. Meissner, Service modelling for the Internet of Things, Federated Conf. Comput. Sci. Inform. Syst. (FedCSIS) 2011 (2011) 949–955.
[28] S. Subasree, N.K. Sakthivel, Design of a new security protocol using hybrid cryptography algorithms||, IJRRAS. 2 (2) (2010) 95–103.
[29] N. Kumar, A secure communication wireless sensor networks through hybrid (AES + ECC) algorithm||, Vol. 386, von LAP LAMBERT Academic Publishing, Koln, Germany, 2012.
[30] W. Ren, Z. Miao, A hybrid encryption algorithm based on DES and RSA in Bluetooth Communication, in: 2010 Second International Conference on Modeling, Simulation and Visualization Methods, 2010, pp. 221-225, doi:10.1109/WMSVM.2010.48.
[31] D.E. Ramaraj, S. Karthikeyan, M. Hemalatha, A design of security protocol using hybrid encryption technique (AES-Rijndael and RSA), Int. J. Comput. Internet Manag. 17 (1) (2009) 78–86.
[32] H. Tange, B. Andersen Attacks and countermeasures on AES and ECC, IN: 2013 16th International Symposium on Wireless Personal Multimedia Communications (WPMC), 2013, pp. 1-5.